

Software Requirements Specification (SRS) - JeevaFlux

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JeevaFlux • Team SYMBIONICS • Smart India Hackathon 2026 • SIH26181

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for JeevaFlux, an on-device AI personal health companion intended to provide continuous monitoring and early warnings while preserving user privacy and supporting operation during connectivity failures.

1.2 Scope

- Physiological and environmental sensor-data acquisition.
- On-device preprocessing, feature extraction and anomaly/risk inference.
- Multimodal sensor fusion and personalized baseline comparison.
- Early-warning generation for heat, environmental exposure, physiological anomalies, fatigue/recovery and selected distress events.
- Mobile companion application for visualization, alerts and user interaction.
- Offline local storage and later synchronization.
- Secure data handling and consent-based sharing.

1.3 Out of Scope

- Autonomous medical diagnosis or replacement of qualified clinical care.
- Guaranteeing prevention of emergencies.
- Final clinical/regulatory claims before formal validation.
- Exact hardware component selection where the deck does not establish a final production BOM.

1.4 Definitions

Term	Meaning
Edge AI	Machine-learning/inference executed locally on the wearable/edge device or nearby phone.
Sensor fusion	Combining multiple physiological, environmental and contextual measurements.
Personal baseline	A user-specific reference derived from normal measurements over time.
Early warning	A risk-oriented notification intended to encourage preventive action before severe deterioration.
Offline-first	Core functions remain usable without continuous network connectivity.
Store-and-forward	Local records are retained and synchronized after connectivity returns.

2. Overall Description

2.1 Product Perspective

JeevaFlux is a distributed health-monitoring system consisting of a sensing device, local intelligence, a mobile companion and optional synchronization/analytics services.

2.2 Users

- Individual users
- Care/support contacts where explicitly enabled
- Authorized institutional users in future pilots

2.3 Assumptions & Dependencies

- Sensors provide readings within their operating specifications.
- The user wears/uses the device correctly.
- The mobile device provides the interfaces required by the selected implementation.
- Connectivity may be unavailable for extended periods; therefore local inference is required.
- Environmental risk data may come from local sensors and/or permitted external data sources where implemented.

3. Functional Requirements

ID	Requirement
JF-REQ-01	Acquire sensor measurements at configured intervals.
JF-REQ-02	Timestamp and associate measurements with device/session context.
JF-REQ-03	Detect invalid, missing or obviously noisy sensor values.
JF-REQ-04	Perform local feature extraction from physiological and environmental streams.
JF-REQ-05	Maintain/update a personalized baseline using an approved adaptation policy.
JF-REQ-06	Fuse multiple signals to estimate contextual health risk.
JF-REQ-07	Generate graded early warnings for supported risk modules.
JF-REQ-08	Display warnings and recommended actions in the mobile companion.
JF-REQ-09	Continue core monitoring and warning functions during network loss.
JF-REQ-10	Store permitted events locally during offline periods.
JF-REQ-11	Synchronize permitted records when connectivity returns.
JF-REQ-12	Allow users to view recent status, trends and event history.
JF-REQ-13	Provide device/connectivity/battery status to the user.
JF-REQ-14	Record alert generation and acknowledgement events for auditability.
JF-REQ-15	Require consent before optional data sharing.

4. Non-Functional Requirements

Area	Requirement / Target
Latency	Time-critical local warnings should be generated with low latency; exact threshold to be established through prototype benchmarking.
Availability	Core warning capability should remain available without

	internet connectivity.
Power	Inference and sensing schedules should be optimized to support continuous/extended operation.
Privacy	Raw sensitive data should remain local unless the user has enabled a justified sharing pathway.
Security	Authenticated access and encrypted synchronization should be used.
Usability	Warnings must be understandable, concise and action-oriented.
Scalability	Architecture should allow additional sensors, risk models and users without redesigning the core pipeline.
Maintainability	Models and firmware/software components should be modular and versioned.
Robustness	The system should tolerate missing/noisy measurements and degraded connectivity.

5. System Interfaces

Interface	Purpose
Physiological sensors	Acquire heart rate, SpO ₂ , skin temperature, motion/activity, sleep/recovery and GSR where implemented.
Environmental sensors	Acquire temperature, humidity and particulate/air-quality measurements where supported.
Mobile app	User status, alerts, history, settings and consent.
Local storage	Offline event and selected measurement storage.
Network	Secure synchronization when available.
Optional external data	Environmental/disaster context sources when integrated and permitted.

6. Data Model

Entity	Representative Fields
User	User ID, baseline profile, consent settings, vulnerability/context preferences.
SensorReading	Timestamp, sensor type, value, quality flag, device/session ID.
EnvironmentalReading	Timestamp, temperature, humidity, particulate/AQI fields where available.
RiskEvent	Timestamp, risk type, severity, contributing features, alert status.
DeviceState	Battery, connectivity, firmware/model version, sensor status.
SyncRecord	Record ID, sync status, timestamp, authorization state.

7. Safety & Validation Requirements

- The product shall clearly distinguish an early-warning aid from a medical diagnosis.
- False alerts and missed alerts must be evaluated during validation.
- Sensor-quality checks shall be incorporated before high-confidence risk decisions.
- Risk thresholds should be calibrated and evaluated across representative user groups.
- Failure modes involving sensor dropout, battery depletion, clock errors and connectivity loss should be tested.

8. Acceptance / Validation Strategy

- Bench-test sensor acquisition and data integrity.
- Evaluate edge inference latency and power consumption.
- Compare risk outputs against labeled test scenarios.
- Measure false-positive and false-negative rates for each supported risk module.
- Conduct controlled usability tests for alert comprehension and response.
- Conduct offline/online transition tests and synchronization integrity checks.